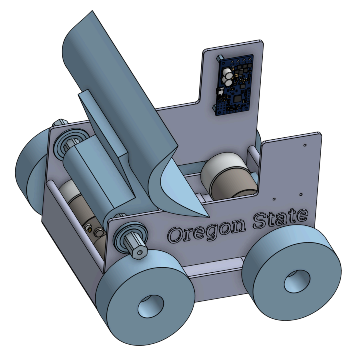
CAD and BOM
June 4 2026 Team # 038 Sumobot! Team Orange

Sumobot



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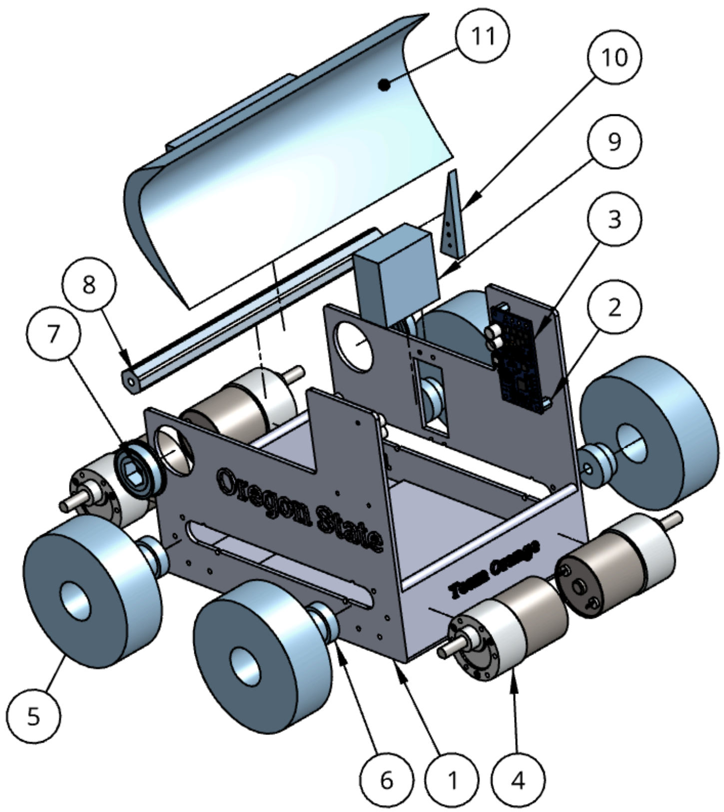
Exploded View of Assembly and Bill of Material

2 1

B B

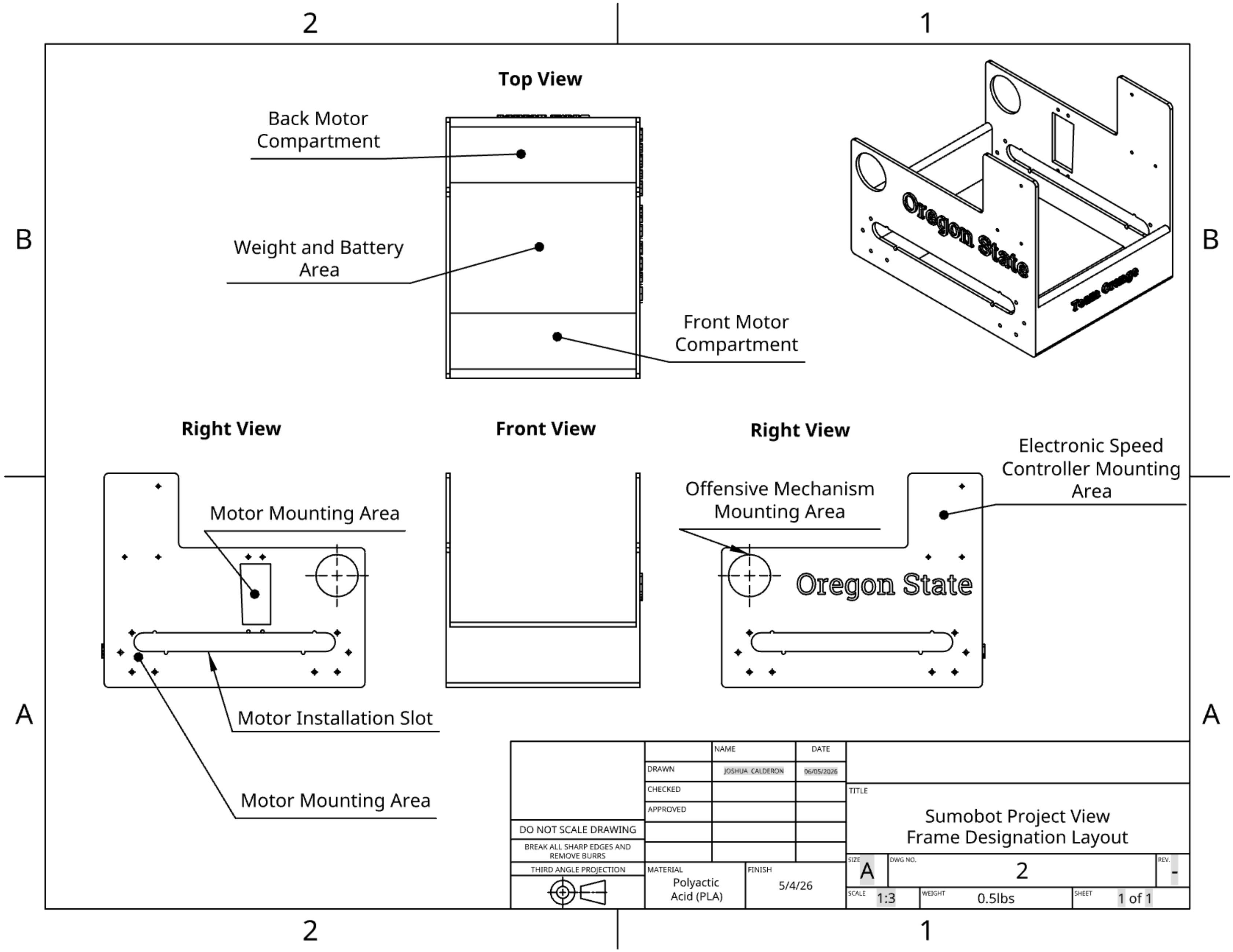


A A

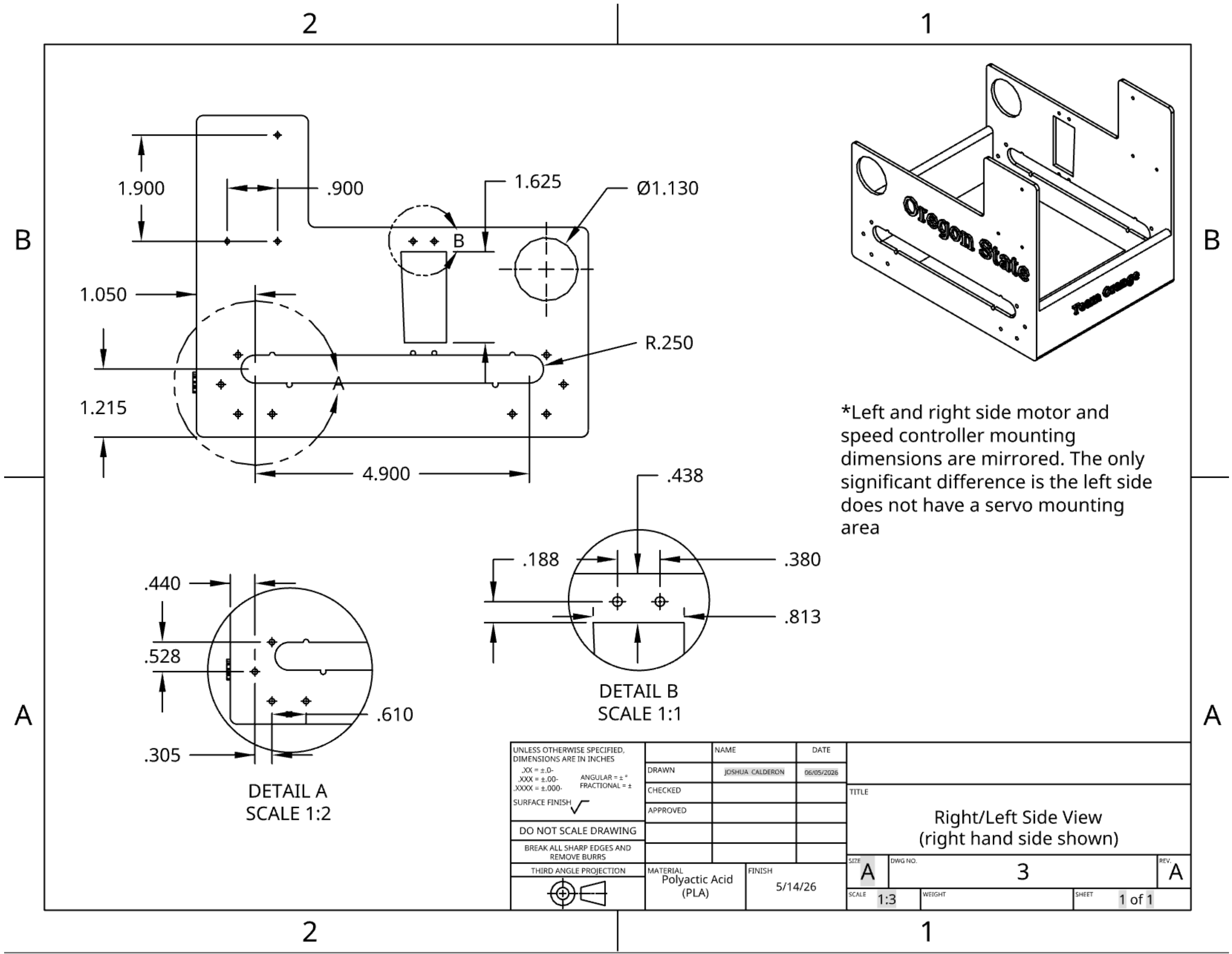
Exploded View and BOM		
June 4, 2026	Joshua Calderon	
Item	**Quantity**	**Description**
1	1	V5 Frame
2	6	1/4" 2-56 Thread Aluminum Stand-offs
3	2	G2 18v15 Motor Controller
4	4	37Dx57L 12V Brushed Motor
5	4	70mmx25mm Wheels
6	4	6mm Diameter Aluminum Wheel Hub
7	2	0.5" Hex Bearing
8	1	6" Long 0.5" Hex Shaft
9	1	SG3003 Servo
10	1	Offensive Mechanism Deployment Arm
11	1	Offensive Mechanism

2 1

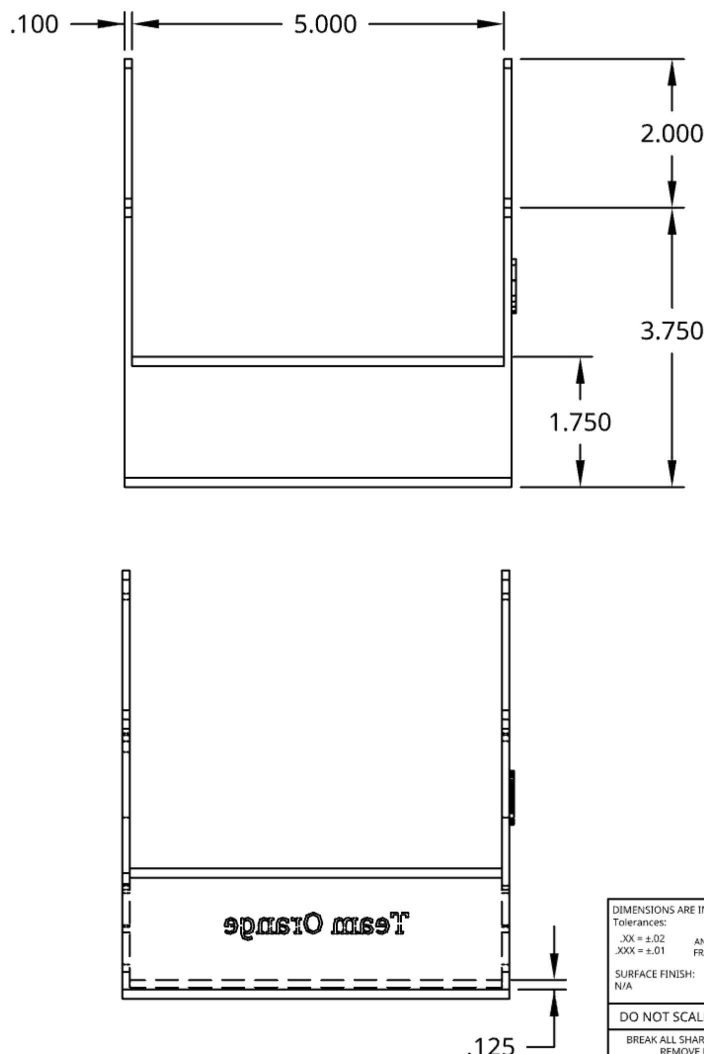
Frame Layout Drawing

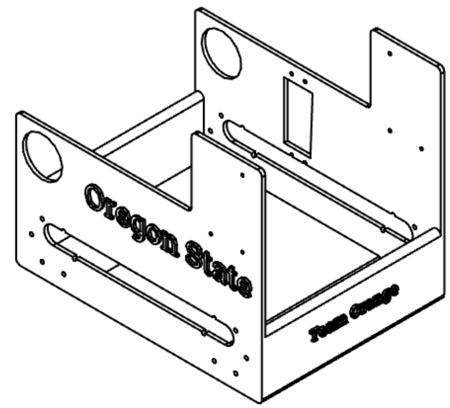


Right/Left View Frame Drawing



Front/Back View Frame Drawing





*Front and back wall dimensions are the same. The only difference is the back side walls have further extrusions upward to allow for mounting of speed controller

DIMENSIONS ARE IN INCHES
Tolerances:
.XX = ±.02
.XXX = ±.01
ANGULAR = ± 1°
FRACTIONAL = ±1/32
SURFACE FINISH:
N/A

DO NOT SCALE DRAWING

BREAK ALL SHARP EDGES AND REMOVE BURRS

THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	JOSHUA CALDERON	06/05/2026
CHECKED		
APPROVED		

MATERIAL: Polyactic Acid (PLA)

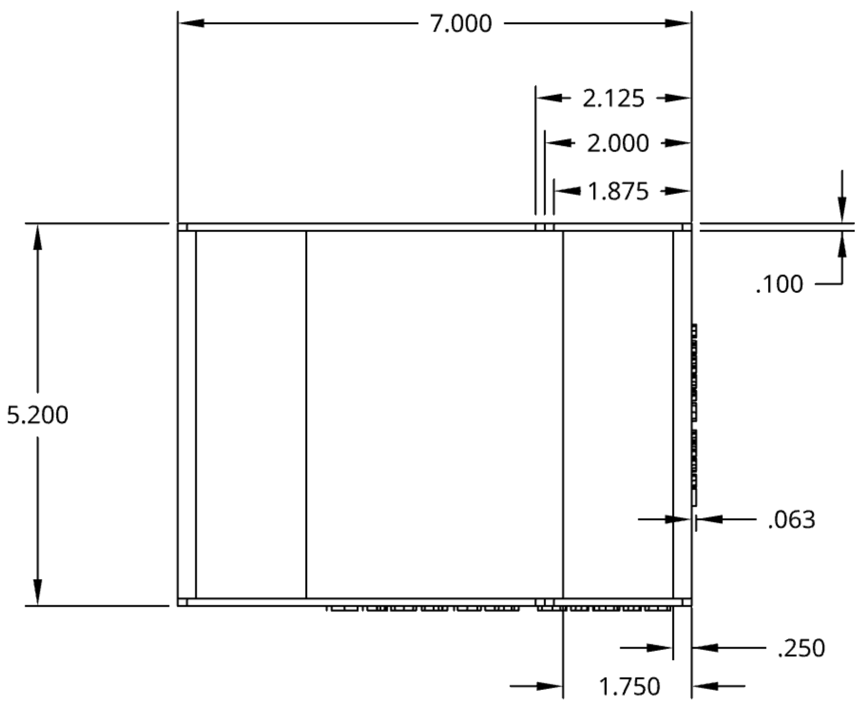
FINISH: 5/4/26

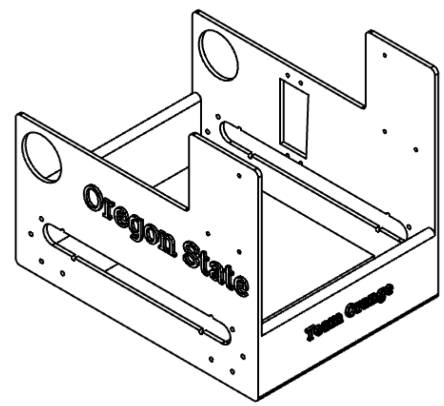
TITLE: Front/Back Wall View (Front wall shown)

SIZE: A | DWG NO.: 4 | REV.: A

SCALE: 1:3 | WEIGHT: 0.5lbs | SHEET: 1 of 1

Top View Frame Drawing





DIMENSIONS ARE IN INCHES
Tolerances:
.XX = ±.02 ANGULAR = ± 1°
.XXX = ±.01 FRACTIONAL = ±1/32
SURFACE FINISH:
N/A

DO NOT SCALE DRAWING

BREAK ALL SHARP EDGES AND REMOVE BURRS

THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	JOSHUA CALDERON	06/07/2026
CHECKED		
APPROVED		

MATERIAL: Polyactic Acid (PLA)

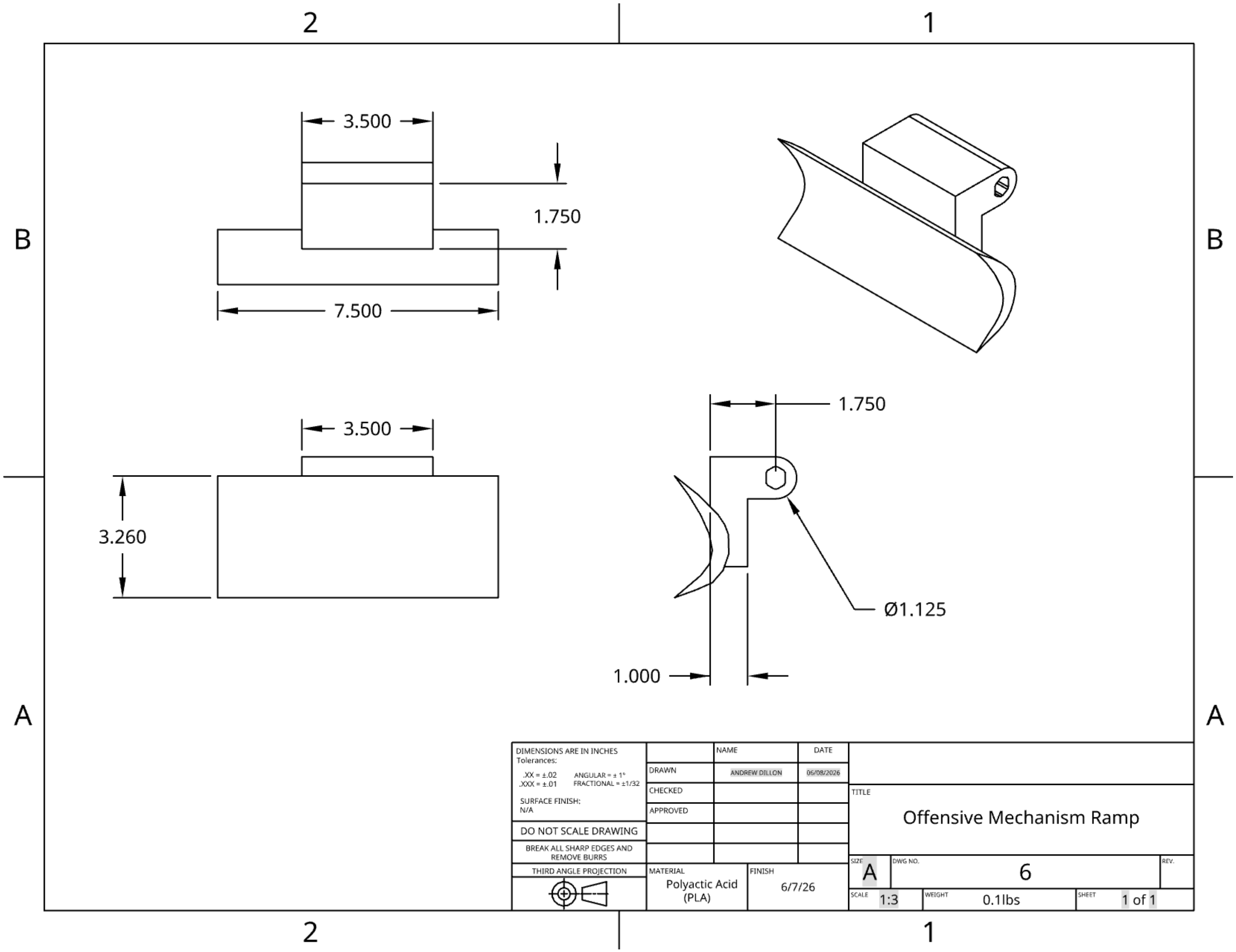
FINISH: 5/7/26

TITLE: Top View

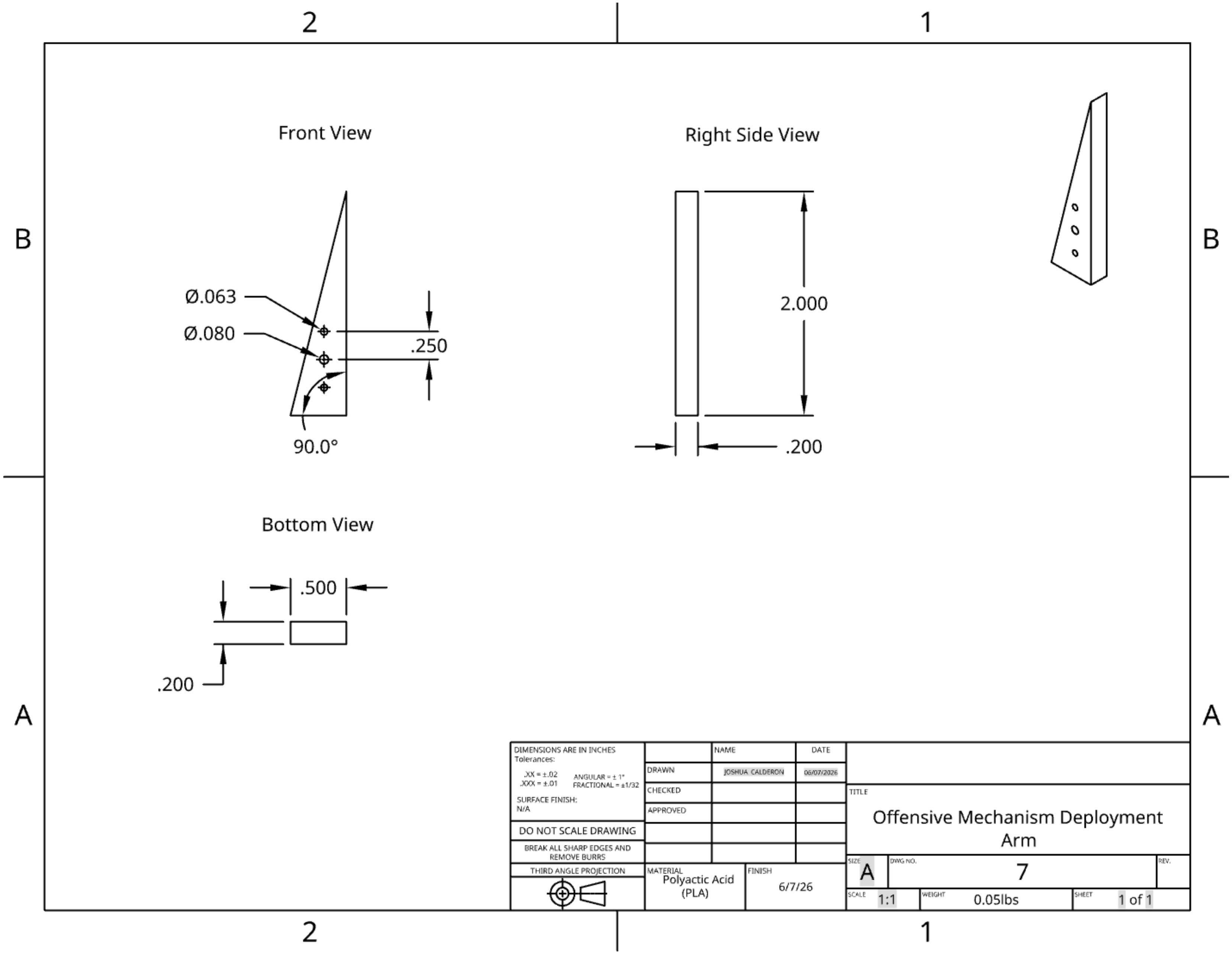
SIZE: A | DWG NO.: 5 | REV.:

SCALE: 1:2 | WEIGHT: 0.5lbs | SHEET: 1 of 1

Offensive Mechanism Drawing



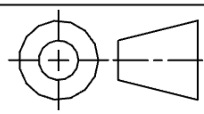
Offensive Mechanism Deployment Arm Drawing



Blown-up view of table (included for better visibility)

DIMENSIONS ARE IN INCHES Tolerances: .XX = ±.02 ANGULAR = ± 1° .XXX = ±.01 FRACTIONAL = ±1/32 SURFACE FINISH: N/A		NAME	DATE	
	DRAWN	JOSHUA CALDERON	06/07/2026	
	CHECKED			TITLE
	APPROVED			Top Vie
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				SIZE A
THIRD ANGLE PROJECTION	MATERIAL Polyactic Acid (PLA)	FINISH 5/7/26		DWG NO. 5

SIZE	A	DWG NO.	5
SCALE	1:2	WEIGHT	0.5lbs



- **<u>Same dimensional specification for both frame and offensive mechanism</u>**
  - **<u>Same Units, Tolerances, Material, Date Made, etc</u>**
    - **<u>Top View drawing used for example</u>**

[SEPARATE BOM LINK]()